

Danh mục: medication

Type 2 diabetes in adults: management

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NICE guideline

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Initial medicines

See the

visual summary

for an overview of the recommendations and additional information to support medicine choice up to the point at which a person starts insulin-based treatment.

Recommendations in this section that cover dipeptidyl peptidase-4 (DPP-4) inhibitors, glucagonlike peptide-1 (GLP-1) receptor agonists, sulfonylureas and sodium–glucose cotransporter-2 (SGLT-2) inhibitors refer to each of these groups of medicines at class level.

For GLP-1 receptor agonists, at the time of publication (February 2026) this only includes liraglutide, dulaglutide, and semaglutide. For subcutaneous semaglutide (Ozempic), this only includes doses up to 1 mg once a week.

SGLT-2 inhibitors and GLP-1 receptor agonists are recommended as much for their cardiovascular and renal benefits as for their glycaemic benefits (unless otherwise specified).

Healthcare professionals should also refer to the summary of product characteristics for individual medicines for contraindications and precautions to take in pregnancy and breastfeeding and for women, trans men and non-binary people of childbearing potential.

1.13 People with type 2 diabetes and no relevant comorbidities

1.13.1

For adults with type 2 diabetes and

no relevant comorbidity

, offer:

modified-release metformin, and

an SGLT-2 inhibitor.

[2026]

1.13.2

If metformin is contraindicated or not tolerated, offer monotherapy with an SGLT-2 inhibitor.

[2026]

Why the committee made these recommendations

Metformin and SGLT-2 inhibitors

There is good evidence that managing type 2 diabetes should aim to improve health holistically (in particular, cardiovascular and renal protection), rather than just aim to meet HbA1c targets.

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The evidence on antidiabetic therapies covered a diverse patient group, including, in varying proportions, people:
with multiple cardiovascular risk factors
with atherosclerotic cardiovascular disease, heart failure or chronic kidney disease
who likely had a lower cardiovascular risk.

The committee agreed that this population likely reflected the population that will be seen most often in general practice. Overall, network and pairwise meta-analyses comparing antidiabetic therapies showed that treatment combining metformin with an SGLT-2-inhibitor was more clinically effective at reducing HbA1c, weight and cardiovascular events than:

any other therapy combining metformin with 1 other medicine, and
metformin alone.

Cardiovascular events covered by the evidence included cardiovascular mortality, myocardial infarction, non-fatal stroke and hospitalisation for heart failure.

The evidence also showed that canagliflozin and dapagliflozin reduced the risk of end-stage renal failure.

Benefits outweigh the risk of adverse events

The committee weighed the greater likelihood of cardiovascular and renal benefits against the risk of volume depletion and genital mycotic infections. The evidence found that, in people taking metformin combined with an SGLT-2 inhibitor, compared with those who did not take an SGLT-2 inhibitor, there were reductions in the number of:

deaths from cardiovascular disease

heart attack or stroke 3-item MACE

hospitalisation for heart failure.

The evidence found, for example, that:

123 out of 1,000 people who did not take an SGLT-2 inhibitor had a major adverse cardiovascular event, known as a 3-item MACE event, over 3 years (that is, a non-fatal myocardial infarction, non-fatal stroke or death from a cardiovascular cause), compared with 107 to 115 people out of 1,000 people who were taking an SGLT-2 inhibitor for 3 years

54 out of 1,000 people who did not take an SGLT-2 inhibitor were hospitalised with heart failure over 3 years, compared with 26 to 40 people out of 1,000 people who were taking an SGLT-2 inhibitor for 3 years.

These clinically important reductions were weighed against the chance that:

between 10 and 100 in 1,000 people would experience genital infections

1 in 1,000 or fewer people would experience rarer events like diabetic ketoacidosis or Fournier's gangrene or other severe infections.

Given this, the committee agreed that, for most people, the benefits outweighed the risks. They agreed that the risk of volume depletion is manageable, and that it would be uncommon for it to lead to diabetic ketoacidosis. They agreed that these risks should be discussed with the person ahead of starting treatment.

Wider access to SGLT-2 inhibitors is likely to reduce health inequalities

The committee also agreed that providing SGLT-2 inhibitors and metformin as a part of standard therapy for most people with type 2 diabetes could reduce inequality. Evidence indicated that interventions to reduce socioeconomic inequalities, including health inequalities, often include actions that address the population or health system level which requires limited voluntary behaviour change on the part of an individual person. Evidence shows that SGLT-2 inhibitors provide net health benefits for people with type 2 diabetes living in the most deprived areas. The committee agreed that this is an important reason for ensuring universal access to SGLT-2 inhibitors.

Choosing a specific medicine

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For the 2026 update, the committee was aware of the large reduction in price of dapagliflozin because generic versions were becoming available. They did not make a recommendation for dapagliflozin, acknowledging that other medicines in the same class were as effective and may become cheaper in the future. However, they support its use while it is the least expensive of the SGLT-2 inhibitors that may be suitable, because it is likely to reduce the cost of implementing the recommendation without negatively affecting quality of care.

People with a relatively lower risk of cardiovascular disease

People with type 2 diabetes have a higher risk of cardiovascular disease than people with the same health-related characteristics and no diabetes. The committee acknowledged that people with type 2 diabetes have an inherent increased lifetime risk of cardiovascular disease before accounting for any other cardiovascular risk factors that are more commonly associated with the condition, such as hypertension and dyslipidaemia. Therefore, in the absence of direct evidence, the committee agreed that even people with a relatively lower cardiovascular risk should be offered an SGLT-2 inhibitor and metformin, because it is important to reduce this risk, including for those without comorbidities or other cardiovascular risk factors. They also noted that, compared with groups who have a higher risk of cardiovascular disease, including people with early onset type 2 diabetes, the population that these recommendations cover is small. This population includes people likely to be 41 to 59 years old, who have no cardiovascular risk factors other than type 2 diabetes (who would otherwise be identified to have a QRISK score below 10%).

Because of their age, they may share similarities with people with early onset type 2 diabetes, for whom earlier intensive treatment is recommended.

Conclusion

Taking the evidence into account, the committee agreed that combining metformin with an SGLT-2 inhibitor is the most clinically effective therapy option, and that this should be the standard initial treatment for people with no relevant comorbidities.

Modified compared to standard-release metformin

There was limited evidence comparing standard-release and modified-release metformin for people with type 2 diabetes and no relevant comorbidities, and the committee agreed that there are benefits to recommending modified-release metformin first. This is because, when compared with standard-release metformin, modified-release metformin:

- has similar clinical effectiveness on HbA1c and weight reduction

- has similar safety results for hypoglycaemia

- is associated, in evidence outside of the protocol for the review, with reductions in gastrointestinal adverse events

- is likely to be better adhered to, and the committee was aware of the downstream costs of non-adherence (for example, cardiovascular and renal adverse events, further appointments and investigations), and

- can fluctuate in price but, in December 2025, cost less than standard-release metformin.

The committee noted that standard-release metformin may be preferable for people with difficulty swallowing because unlike modified-release metformin, it can be crushed and is available in a liquid form. This may be appropriate for people with dementia or with learning disabilities in whom dysphagia is more common. The committee did not make a recommendation about this because they were aware that

NEWT guidelines

provide further information to support decision making for people with swallowing difficulties.

The committee agreed that this should be addressed in a conversation between the person and the healthcare professional when prescribing the medication.

GLP-1 receptor agonists and tirzepatide

GLP-1 receptor agonists and tirzepatide were not found to be or did not have evidence for clinical or cost effectiveness

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for this population, so were not recommended.

Medicines for people who cannot take metformin

The evidence showed that SGLT-2 inhibitors reduced cardiovascular events compared with placebo when metformin is the background therapy. The committee agreed that even though the evidence was limited for people with no relevant comorbidities, this benefit would also be seen in people for whom metformin is contraindicated. Therefore, because the committee wanted people with type 2 diabetes to continue to gain the cardiovascular benefits seen in the evidence, they recommended monotherapy with an SGLT-2 inhibitor when metformin is contraindicated.

Recommendation for research

The committee reviewed real-world evidence that SGLT-2 inhibitors are under-prescribed, particularly to women and older people, people from some ethnic backgrounds, and people who have experienced higher levels of deprivation when sex and age are accounted for. They agreed that further research is needed to understand the reasons behind this so made a

recommendation for research on improving access to SGLT-2 inhibitors

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendations might affect practice

Metformin

The recommendations may lead to a change in current practice but should not lead to a significant cost or resource impact. The price of modified-release metformin can fluctuate but, in December 2025, was lower than the cost of standard-release metformin.

SGLT-2 inhibitors

SGLT-2 inhibitors were recommended by NICE in 2022 for:

all people with chronic heart failure

all people with atherosclerotic cardiovascular disease

some people at high risk of developing cardiovascular disease.

Recommendations for SGLT-2 inhibitors for people with chronic kidney disease were previously made through different guidance but are not expected to reflect a significant change in practice.

However, real-world evidence (2026) shows that SGLT-2 inhibitors are under-prescribed throughout the UK. The 2026 recommendations may increase the number of people who are offered SGLT-2 inhibitors, which will increase prescribing costs. But broader access to SGLT-2 inhibitors may also result in long-term medicine costs being partially offset by fewer people needing treatment for atherosclerotic cardiovascular disease.

Based on these recommendations, people at lower risk of developing cardiovascular disease, who previously would not have had access to SGLT-2 inhibitors, can now access them. The committee did not believe this to be a large group and so the impact of this is likely to be minimal.

Number of appointments related to these medicines

The recommendations may lead to an increase in the number of appointments required to optimise the medications being added. However, this can be managed and may lead to reductions in the long term because:

SGLT-2 inhibitors can be prescribed at the same time as metformin, with a plan for starting medicines sequentially the cardiovascular and renal benefits of SGLT-2 inhibitors will reduce the number of appointments needed to treat atherosclerotic cardiovascular disease, established heart failure and chronic kidney disease.

1.14 People with heart failure (with any ejection fraction, unless specified)

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1.14.1

For adults with type 2 diabetes and heart failure offer:

modified-release metformin, and

an SGLT-2 inhibitor.

[2026]

1.14.2

If metformin is contraindicated or not tolerated, offer monotherapy with an SGLT-2 inhibitor.

[2026]

Why the committee made these recommendations

Metformin and SGLT-2 inhibitors

There is a significant body of evidence showing that type 2 diabetes management should aim at holistic health improvements (in particular, cardiovascular and renal protection), rather than just HbA1c targets.

Overall, network and pairwise meta-analyses comparing antidiabetic therapies showed that therapy combining metformin with an SGLT-2 inhibitor was more clinically effective at reducing HbA1c, weight and cardiovascular events than:

any other therapy combining metformin with 1 other medicine, and

metformin alone.

Cardiovascular events covered included cardiovascular mortality, myocardial infarction, non-fatal stroke and hospitalisation for heart failure. Evidence showed that canagliflozin and dapagliflozin also reduced the risk of end-stage renal failure.

The committee agreed that therapy combining metformin with an SGLT-2 inhibitor is the most clinically effective option for people with heart failure and recommended this as the standard initial treatment.

There was limited evidence for people with type 2 diabetes and heart failure. However, for people with type 2 diabetes who are at high risk of developing cardiovascular disease, there was strong evidence that SGLT-2 inhibitors reduced the number of hospitalisations due to heart failure.

During the 2026 update of the guideline, the committee was aware of the large reduction in price of dapagliflozin because generic versions of the medicine were becoming available. They did not make a recommendation for this specific medicine, acknowledging that other medicines in the same class were as effective and may become cheaper in the future. However, they support its use while it is the least expensive of the SGLT-2 inhibitors that may be suitable, because it is likely to reduce the cost of implementing the recommendation without impacting the quality of care for most people with type 2 diabetes.

Modified compared to standard-release metformin

There was limited evidence comparing standard-release and modified-release metformin. On weighing up the evidence, the committee agreed that there were benefits to recommending modified-release metformin first. This was because, when compared with standard-release metformin, modified-release metformin:

has similar clinical effectiveness on HbA1c and weight reduction

has similar safety results for hypoglycaemia

was associated, in evidence outside of the protocol for the review, with reductions in gastrointestinal adverse events

is likely to be better adhered to, and the committee was aware of the downstream costs of non-adherence (for example, cardiovascular and renal adverse events, further appointments and investigations)

can fluctuate in price but, in December 2025, cost less than standard-release metformin.

The committee noted that standard-release metformin can be preferable for people with difficulty swallowing because it

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can be crushed and is available in a liquid form while modified-release metformin cannot. This may be more useful for some people with dementia and some people with learning disabilities in whom dysphagia is more common. The committee did not make a recommendation about this because they were aware that

NEWT guidelines

provide further information to support decision making for people with swallowing difficulties.

The committee agreed that this should be addressed in a conversation between the person and the healthcare professional when prescribing the medication.

Medicines for people who cannot take metformin

The trial evidence showed that SGLT-2 inhibitors reduced cardiovascular events compared with placebo when metformin was the background therapy. The committee agreed that this benefit of using SGLT-2 inhibitors would also be seen in people with contraindications to metformin, even though the evidence was limited for this group. Therefore, as the committee wanted people with type 2 diabetes to continue to gain the cardiovascular benefits seen in the evidence, they recommended that when people have a contraindication to metformin, they should have monotherapy with an SGLT-2 inhibitor.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendations might affect practice

Metformin

The recommendations about the use of metformin may lead to a change in current practice but should not lead to a significant cost or resource impact. The price of modified-release metformin can fluctuate but, in December 2025, was lower than the cost of standard-release metformin.

SGLT-2 inhibitors

SGLT-2 inhibitors were recommended by NICE in 2022. They were recommended for:

all people with chronic heart failure

some people at high risk of developing cardiovascular disease.

However, real-world evidence shows that SGLT-2 inhibitors are under-prescribed throughout the UK. The recommendations may increase the number of people who are offered SGLT-2 inhibitors, which will increase prescribing costs. But the long-term medicine cost associated with broader access to SGLT-2 inhibitors may be partially offset if it results in fewer people needing treatment for atherosclerotic cardiovascular disease or hospitalisation for heart failure.

Number of appointments related to these medicines

The recommendations may lead to an increase in the number of appointments required to optimise the medications being added. The committee believes this can be managed and will lead to reductions long term because:

SGLT-2 inhibitors can be prescribed at the same time as metformin, with a plan for starting medicines sequentially the cardiovascular and renal benefits of SGLT-2 inhibitors will reduce the number of appointments needed to treat atherosclerotic cardiovascular disease, established heart failure and chronic kidney disease.

1.15 People with atherosclerotic cardiovascular disease

1.15.1

For adults with type 2 diabetes and

atherosclerotic cardiovascular disease

, offer:

modified-release metformin, and

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an SGLT-2 inhibitor, and

subcutaneous semaglutide (Ozempic), up to 1 mg once a week, for its cardiovascular, renal and glycaemic benefits.

[2026]

1.15.2

If metformin is contraindicated or not tolerated, offer:

an SGLT-2 inhibitor, and

subcutaneous semaglutide (Ozempic), up to 1 mg once a week, for its cardiovascular, renal and glycaemic benefits.

[2026]

Why the committee made these recommendations

Metformin and SGLT-2 inhibitors

There is a significant body of evidence showing that type 2 diabetes management should aim at holistic health improvements (in particular, cardiovascular and renal protection), rather than just HbA1c targets.

Overall, network and pairwise meta-analyses comparing antidiabetic therapies showed that therapy combining metformin with an SGLT-2 inhibitor was more clinically effective at reducing HbA1c, weight and cardiovascular events than:

any other therapy combining metformin with 1 other medicine, and

metformin alone.

Cardiovascular events covered included cardiovascular mortality, myocardial infarction, non-fatal stroke and hospitalisation for heart failure. Evidence showed that canagliflozin and dapagliflozin reduced the risk of end-stage renal failure.

The committee agreed that therapy combining metformin with an SGLT-2 inhibitor is the most clinically effective option for people with atherosclerotic cardiovascular disease and recommended this as the standard initial treatment.

Data on health inequalities showed that people living in the most deprived areas experience the greatest benefits for their health from SGLT-2 inhibitors. The committee believe this is an important reason for ensuring universal access to SGLT-2 inhibitors.

During the 2026 update of the guideline, the committee was aware of the large reduction in price of dapagliflozin because generic versions of the medicine were becoming available. They did not make a recommendation for this specific medicine, acknowledging that other medicines in the same class were as effective and may become cheaper in the future. However, they support its use while it is the least expensive of the SGLT-2 inhibitors that may be suitable, because it is likely to reduce the cost of implementing the recommendation without impacting the quality of care for most people with type 2 diabetes.

Modified compared to standard-release metformin

There was limited evidence comparing standard-release and modified-release metformin. On weighing up the evidence, the committee agreed that there were benefits to recommending modified-release metformin first. This was because, when compared with standard-release metformin, modified-release metformin:

has similar clinical effectiveness on HbA1c and weight reduction

has similar safety results for hypoglycaemia

was associated, in evidence outside of the protocol for the review, with reductions in gastrointestinal adverse events

is likely to be better adhered to, and the committee was aware the downstream costs of non-adherence (for example: cardiovascular and renal adverse events, further appointments and investigations)

can fluctuate in price but, in December 2025, cost less than standard-release metformin.

The committee noted that standard-release metformin can be preferable for people with difficulty swallowing because it

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can be crushed and is available in a liquid form while modified-release metformin cannot. This may be more useful for some people with dementia and some people with learning disabilities in whom dysphagia is more common. The committee did not make a recommendation about this because they were aware that the

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provide further information to support decision making for people with swallowing difficulties.

The committee agreed that this should be addressed in a conversation between the person and the healthcare professional when prescribing the medication.

Medicines for people who cannot take metformin

The trial evidence showed that SGLT-2 inhibitors reduced cardiovascular events compared with placebo when metformin was the background therapy. The committee agreed that this benefit of using SGLT-2 inhibitors would also be seen in people with contraindications to metformin, even though the evidence was limited for this group. Therefore, as the committee wanted people with type 2 diabetes to continue to gain the cardiovascular benefits seen in the evidence, they recommended that when people have a contraindication to metformin, they should have monotherapy with an SGLT-2 inhibitor.

Subcutaneous semaglutide

Evidence showed that subcutaneous semaglutide (Ozempic) was the most cost-effective GLP-1 receptor agonist. This evidence was based on clinical benefits identified in a population that included a large proportion of people with atherosclerotic cardiovascular disease but also some people who did not have atherosclerotic cardiovascular disease but were at high risk of it (because they had type 2 diabetes). Evidence was only identified for doses up to 1 mg of subcutaneous semaglutide once a week. The committee agreed that evidence from this mixed population indirectly applied because of the large proportion of people in the group who did have atherosclerotic cardiovascular disease.

Subcutaneous semaglutide (Ozempic), up to 1 mg once a week, also had the best clinical results. Evidence showed that it made a clinically important reduction to the person's:

risk of major adverse cardiovascular events and

HbA1c and

weight.

The results for these outcomes were precise, which means that there is very little uncertainty about them. Other GLP-1 receptor agonists did not achieve this. Liraglutide was cost effective in a sensitivity analysis taking into account projected price reductions after the generic version became available. But the committee agreed that it was not sufficiently clinically effective to justify adding it to the recommendation. In addition, evidence was not available to the committee to evaluate tirzepatide for this population at this time. As a result, the committee agreed to specifically recommend subcutaneous semaglutide (Ozempic), up to 1 mg once a week, rather than any other GLP-1 receptor agonist or tirzepatide.

The committee noted that there are potential serious side effects with GLP-1 receptor agonists and tirzepatide, and a potential for misuse. However, they did not make a recommendation on monitoring because it is covered by MHRA drug safety updates. See MHRA guidance on

GLP-1 receptor agonists: reminder of the potential side effects and to be aware of the potential for misuse

GLP-1 receptor agonists and dual GLP-1/GIP receptor agonists: strengthened warnings on acute pancreatitis, including necrotising and fatal cases

semaglutide (Wegovy, Ozempic and Rybelsus): risk of non-arteritic anterior ischemic optic neuropathy (NAION)

for more details.

The committee made the decision to recommend this medicine in combination with metformin and an SGLT-2 inhibitor

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based on:

evidence they had from a pooled analysis

their own clinical experience.

The evidence in the pooled network meta-analysis came from a review that looked at the cost and clinical effectiveness of adding subsequent therapies to previous treatment. It showed clinical benefits from GLP-1 receptor agonists, but most studies in the evidence review did not give separate results based on the number or type of other treatments received. A small number of studies specifically included triple therapy combining GLP-1 receptor agonists, SGLT-2 inhibitors and metformin. When compared in health economic evaluation, adding subcutaneous semaglutide to an SGLT-2 inhibitor and metformin was cost effective.

The evidence for combination therapy with metformin and SGLT-2 inhibitors showed that the cardiovascular benefits came from the SGLT-2 inhibitors alone. This was clear because people receiving metformin and placebo did not get the same benefits. When compared with placebo in clinical trials, GLP-1 receptor agonists, including semaglutide also showed cardiovascular benefits, regardless of other treatment received. Because of this, the committee agreed that people with atherosclerotic cardiovascular disease should receive therapy combining an SGLT-2 inhibitor and subcutaneous semaglutide if metformin is contraindicated or not tolerated.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendations might affect practice

Metformin

The recommendations about the use of metformin may lead to a change in current practice but should not lead to a significant cost or resource impact. The price of modified-release metformin can fluctuate but, in December 2025, was lower than the cost of standard-release metformin.

SGLT-2 inhibitors were recommended by NICE in 2022. They were recommended for:

all people with atherosclerotic cardiovascular disease

some people at high risk of developing cardiovascular disease.

However, real-world evidence shows that SGLT-2 inhibitors are under-prescribed throughout the UK. The recommendations may increase the number of people who are offered SGLT-2 inhibitors, which will increase prescribing costs. But broader access to SGLT-2 inhibitors may also result in long-term medicine costs being partially offset by fewer people needing treatment for atherosclerotic cardiovascular disease.

Subcutaneous semaglutide

Subcutaneous semaglutide is a GLP-1 receptor agonist. GLP-1 receptor agonists were previously reserved for later treatment phases. Recommending subcutaneous semaglutide (Ozempic), up to 1 mg once a week, for some people as part of initial therapy will increase costs. Taking subcutaneous semaglutide will mean that DPP-4 inhibitor use will reduce. Early intervention could lead to weight loss as a beneficial side effect that may result in better long-term prognoses. If maintained, this will reduce needs for both long-term treatment and later stage treatments (such as insulin).

Number of appointments related to these medicines

The recommendations may lead to an increase in the number of appointments required to optimise the medications being added. The committee believes this can be managed and will lead to reductions in the long term because:

SGLT-2 inhibitors can be prescribed at the same time as metformin, with a plan for starting medicines sequentially

the cardiovascular and renal benefits of SGLT-2 inhibitors will reduce the number of appointments needed to treat

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atherosclerotic cardiovascular disease, established heart failure and chronic kidney disease

the cardiovascular benefits of subcutaneous semaglutide will reduce the number of appointments needed to treat atherosclerotic cardiovascular disease and established heart failure.

1.16 People with early onset type 2 diabetes

1.16.1

For adults with

early onset type 2 diabetes

, offer modified-release metformin and an SGLT-2 inhibitor, and consider adding either:

a GLP-1 receptor agonist for its cardiovascular, renal and glycaemic benefits or

tirzepatide for its glycaemic benefits.

[2026]

1.16.2

If metformin is contraindicated or not tolerated, offer an SGLT-2 inhibitor and consider adding either:

a GLP-1 receptor agonist for its cardiovascular, renal and glycaemic benefits or

tirzepatide for its glycaemic benefits.

[2026]

Why the committee made these recommendations

Metformin and SGLT-2 inhibitors

There is good evidence that managing type 2 diabetes should aim to improve health holistically (in particular, cardiovascular and renal protection), rather than just HbA1c. Based on their experience, the committee agreed that people with early onset type 2 diabetes:

have a very high lifetime risk of cardiovascular and renal complications, and of dying from them, and are more likely to be living with obesity.

Early intensive treatment can provide benefits by preventing these future negative outcomes.

Though there were no clinical trials focusing solely on people with early onset diabetes, there was good evidence in the wider population of people with type 2 diabetes, which the committee agreed could be extrapolated to people with early onset type 2 diabetes. It showed that:

both SGLT-2 inhibitors and GLP-1 receptor agonists reduce the risk of cardiovascular events (cardiovascular mortality, myocardial infarctions, non-fatal strokes and hospitalisation for heart failure) and of developing end-stage renal disease. SGLT-2 inhibitors, GLP-1 receptor agonists and tirzepatide reduce HbA1c, and lead to weight loss as a beneficial side effect.

The committee agreed that the benefits could be increased with long-term use.

The health economic evidence for this population was highly uncertain. The committee concluded that this was because: there were no clinical trials focusing solely on early onset diabetes,

the trial informing the model only included a small number of people with early onset type 2 diabetes, and

the short time horizon for evaluating treatment benefits may have underestimated long-term advantages for this group.

Data on health inequalities showed that people living in the most deprived areas experience the greatest health benefits from SGLT-2 inhibitors. The committee agreed that this is an important reason for ensuring universal access to SGLT-2 inhibitors.

During the 2026 update of the guideline, the committee was aware of the large reduction in price of dapagliflozin because generic versions of the medicine were becoming available. They did not make a recommendation for this specific medicine, acknowledging that other medicines in the same class were as effective and may become cheaper in

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the future. However, they support its use while it is the least expensive of the SGLT-2 inhibitors that may be suitable, because it is likely to reduce the cost of implementing the recommendation without impacting the quality of care for most people with type 2 diabetes.

Modified-release compared to standard-release metformin

There was limited evidence comparing standard-release and modified-release metformin, and the committee agreed that there are benefits to recommending modified-release metformin first. This is because, when compared with standard-release metformin, modified-release metformin:

- has similar clinical effectiveness on HbA1c and weight reduction

- has similar safety results for hypoglycaemia

- is associated, in evidence outside of the protocol for the review, with reductions in gastrointestinal adverse events

- is likely to be better adhered to, and the committee was aware of the downstream costs of non-adherence (for example, cardiovascular and renal adverse events, further appointments and investigations), and

- can fluctuate in price but, in December 2025, cost less than standard-release metformin.

The committee noted that standard-release metformin may be preferable for people with difficulty swallowing because unlike modified-release metformin, it can be crushed and is available in a liquid form. This may be appropriate for people with dementia or learning disabilities in whom dysphagia is more common. The committee did not make a recommendation about this because they were aware that the

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provide further information to support decision making for people with swallowing difficulties.

The committee agreed that this should be addressed in a conversation between the person and the healthcare professional when prescribing the medication.

GLP-1 receptor agonists and tirzepatide

The committee recommended combining a GLP-1 agonist or tirzepatide with metformin and an SGLT-2 inhibitor based on evidence from a pooled network meta-analysis and their clinical experience.

The evidence in the pooled network meta-analysis used in the health economic modelling came from a review of people at higher risk of developing cardiovascular disease or people with existing atherosclerotic cardiovascular disease adding subsequent therapies to previous treatment. It showed benefits from GLP-1 receptor agonists, but most studies did not give separate results based on the number or type of other treatments received. A small number of studies included triple therapy combining GLP-1 receptor agonists, SGLT-2 inhibitors and metformin. When compared in health economic evaluation, adding most GLP-1 receptor agonists was not cost effective, while adding liraglutide to an SGLT-2 inhibitor and metformin reported an incremental cost-effectiveness ratio (ICER) approaching £20,000 per quality-adjusted life year (QALY) gained. Tirzepatide was not analysed for this population.

The evidence for combination therapy with metformin and SGLT-2 inhibitors showed that the cardiovascular benefits came from the SGLT-2 inhibitors alone. This was clear because the people receiving metformin and placebo did not get the same benefits. When compared with placebo in clinical trials, GLP-1 receptor agonists also showed cardiovascular benefits regardless of other treatment received. The evidence evaluated for tirzepatide did not show cardiovascular benefits, which leaves some uncertainty about its use for this purpose. However, the committee acknowledged the benefits in reducing HbA1c and weight, and how this could lead to beneficial cardiovascular outcomes in the long term.

The committee agreed that GLP-1 receptor agonists and tirzepatide should be considered in addition to metformin and SGLT-2 inhibitors, given the:

- relatively small size of this group

- health inequalities that this group would face if they did not receive treatment early, and

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challenges in identifying appropriate data.

The committee also made a

recommendation for research on treatments for people with early onset diabetes

The committee noted potential serious side effects with GLP-1 receptor agonists and tirzepatide, and a potential for misuse. However, they did not make a recommendation for monitoring because it is covered by MHRA drug safety updates. See MHRA guidance on

GLP-1 receptor agonists: reminder of the potential side effects and to be aware of the potential for misuse

GLP-1 receptor agonists and dual GLP-1/GIP receptor agonists: strengthened warnings on acute pancreatitis, including necrotising and fatal cases

semaglutide (Wegovy, Ozempic and Rybelsus): risk of non-arteritic anterior ischemic optic neuropathy (NAION)

for more details.

Medicines for people who cannot take metformin

The evidence showed that SGLT-2 inhibitors reduce cardiovascular events compared with placebo when metformin is the background therapy. The committee agreed that even though the evidence was limited for this group, this benefit would also be seen in people for whom metformin is contraindicated. Therefore, given that the committee wanted people with type 2 diabetes to continue to gain the cardiovascular benefits seen in the evidence, they recommended monotherapy with an SGLT-2 inhibitor when metformin is contraindicated.

They agreed that people with early onset diabetes should receive an SGLT-2 inhibitor, and that adding a GLP-1 receptor agonist or tirzepatide could be considered if metformin is contraindicated or not tolerated.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendations might affect practice

Metformin

The recommendations may lead to a change in current practice but should not lead to a significant cost or resource impact. The price of modified-release metformin can fluctuate but, in December 2025, was lower than the cost of standard-release metformin.

SGLT-2 inhibitors

SGLT-2 inhibitors were recommended by NICE in 2022 for some people at high risk of developing cardiovascular disease.

However, real-world evidence shows that SGLT-2 inhibitors are under-prescribed throughout the UK. The recommendations may increase the number of people who are offered SGLT-2 inhibitors, which will increase prescribing costs. But broader access to SGLT-2 inhibitors may also result in long-term medicine costs being partially offset by fewer people needing treatment for atherosclerotic cardiovascular disease, especially in this population, when taking a long-term perspective.

GLP-1 receptor agonists and tirzepatide

GLP-1 receptor agonists and tirzepatide were previously reserved for later treatment phases. Recommending these for some people as part of initial therapy will increase costs. Increased GLP-1 receptor agonists or tirzepatide use will reduce DPP-4 inhibitor use. Early intervention could lead to weight loss as a beneficial side effect that may result in better long-term prognoses. If maintained, this will reduce needs for both long-term treatment and later stage treatments (such as insulin).

Number of appointments related to these medicines

Danh mục: medication

The recommendations may lead to an increase in the number of appointments required to optimise the medications being added. However, this can be managed and may lead to reductions in the long term because:

SGLT-2 inhibitors can be prescribed at the same time as metformin, with a plan for starting medicines sequentially the cardiovascular and renal benefits of SGLT-2 inhibitors will reduce the number of appointments needed to treat atherosclerotic cardiovascular disease, established heart failure and chronic kidney disease

the cardiovascular benefits of GLP-1 receptor agonists will reduce the number of appointments needed to treat atherosclerotic cardiovascular disease and established heart failure.

1.17 People living with obesity

1.17.1

For adults with type 2 diabetes who are living with obesity, offer:

modified-release metformin and

an SGLT-2 inhibitor.

For a definition of obesity, see the

section on classifying overweight and obesity in adults in NICE's guideline on overweight and obesity management [2026]

1.17.2

If metformin is contraindicated or not tolerated, offer monotherapy with an SGLT-2 inhibitor.

[2026]

Why the committee made these recommendations

The evidence comparing antidiabetic therapies for people with no relevant comorbidities included people living with obesity. However, in most studies it was not possible to separate out this group from the larger study population and identify specific effects for people living with obesity. Given the limitations of the evidence, the committee recommended the same medicines for this group as for other people with type 2 diabetes and no other specific comorbidities.

The committee recommended these therapies for people living with obesity because of their glycaemic reduction properties, and cardiovascular and renal benefits. Weight reduction may be a side effect of some medicines (including GLP-1 receptor agonists and tirzepatide, SGLT-2 inhibitors, and metformin), which may be important to the person with type 2 diabetes and their healthcare professionals. However, if weight reduction is the primary aim of the treatment, the committee agreed that healthcare professionals should follow NICE's guideline on overweight and obesity management instead of this guideline.

For the wider population, therapy with metformin and an SGLT-2 inhibitor was more clinically effective at reducing HbA1c, weight and cardiovascular events than:

any other therapy combining metformin with 1 other medicine, and metformin alone.

Cardiovascular events covered included cardiovascular mortality, myocardial infarction, non-fatal stroke and hospitalisation for heart failure. Evidence showed that canagliflozin and dapagliflozin reduced the risk of end-stage renal failure.

The committee agreed that therapy combining metformin with an SGLT-2 inhibitor is the most clinically effective option for people living with obesity and recommended this as the standard initial treatment.

SGLT-2 inhibitors were cost effective for people living with obesity. Data on health inequalities also showed that people living in the most deprived areas experience the greatest benefits for their health from SGLT-2 inhibitors. The committee believe this is an important reason for ensuring universal access to SGLT-2 inhibitors.

During the 2026 update of the guideline, the committee was aware of the large reduction in price of dapagliflozin

Danh mục: medication

because generic versions of the medicine were becoming available. They did not make a recommendation for this specific medicine, acknowledging that other medicines in the same class were as effective and may become cheaper in the future. However, they support its use while it is the least expensive of the SGLT-2 inhibitors that may be suitable, because it is likely to reduce the cost of implementing the recommendation without impacting the quality of care for most people with type 2 diabetes.

GLP-1 receptor agonists were not cost effective in the health economic modelling for this population. Compared to people with early onset type 2 diabetes, the lifetime risk of cardiovascular disease is lower. Bearing in mind the cardiovascular protection already being provided by SGLT-2 inhibitors and metformin, GLP-1 receptor agonists and tirzepatide were not recommended as initial medicines and were instead recommended as treatment options if further medicines are needed.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendations might affect practice

Metformin

The recommendations about the use of metformin may lead to a change in current practice but should not lead to a significant cost or resource impact. The price of modified-release metformin can fluctuate but, in December 2025, was lower than the cost of standard-release metformin.

SGLT-2 inhibitors

SGLT-2 inhibitors were recommended by NICE in 2022. They were recommended for some people at high risk of developing cardiovascular disease.

However, real-world evidence shows that SGLT-2 inhibitors are under-prescribed throughout the UK. The recommendations may increase the number of people who are offered SGLT-2 inhibitors, which will increase prescribing costs. But broader access to SGLT-2 inhibitors may also result in long-term medicine costs being partially offset by fewer people needing treatment for atherosclerotic cardiovascular disease.

Number of appointments related to these medicines

The recommendations may lead to an increase in the number of appointments required to optimise the medications being added. The committee believes this can be managed and will lead to reductions in the long term because:

SGLT-2 inhibitors can be prescribed at the same time as metformin, with a plan for starting medicines sequentially the cardiovascular and renal benefits of SGLT-2 inhibitors will reduce the number of appointments needed to treat atherosclerotic cardiovascular disease, established heart failure and chronic kidney disease.

1.18 People with chronic kidney disease

Medicines vary in their contraindications and precautions for use in people with renal impairment. See

NICE's information on prescribing medicines

and refer to the summary of product characteristics for individual products.

1.18.1

For adults with type 2 diabetes and an estimated glomerular filtration rate (eGFR) above 30 ml/min/1.73 m

Offer modified-release metformin and an SGLT-2 inhibitor.

If metformin is contraindicated or not tolerated, offer monotherapy with an SGLT-2 inhibitor.

[2026]

1.18.2

For adults with type 2 diabetes and an eGFR of 20 ml/min/1.73 m

Danh mục: medication

and up to 30 ml/min/1.73 m

, offer:

either dapagliflozin or empagliflozin and
a DPP-4 inhibitor.

[2026]

1.18.3

For adults with type 2 diabetes and an eGFR below 20 ml/min/1.73 m

, consider a DPP-4 inhibitor.

[2026]

1.18.4

If a DPP-4 inhibitor is contraindicated, not tolerated or not effective, consider:

pioglitazone or

an insulin-based treatment.

[2026]

1.18.5

For guidance on managing other aspects of kidney disease in adults with type 2 diabetes, see

NICE's guideline on chronic kidney disease

[2015]

Why the committee made these recommendations

eGFR above 30 ml/min/1.73 m

Little evidence was identified specifically for people with chronic kidney disease. However, the committee agreed that people with an eGFR above 30 ml/min/1.73 m

should see the same benefits from diabetes medicines as people without chronic kidney disease. In the health economic analysis, metformin and SGLT-2 inhibitors were less costly and more effective than other medicines. The committee noted that while the cardiovascular and renal protection provided by SGLT-2 inhibitors are retained at eGFR values below 45 ml/min/1.73 m

, the glycaemic benefits may reduce. Glycaemic benefits can come from metformin, but there may be a need to add further therapy to provide these. Options for this are provided in the section on treatment options if further medicines are needed.

Data on health inequalities showed that people living in the most deprived areas experience the greatest benefits for their health from SGLT-2 inhibitors. The committee believe this is an important reason for ensuring universal access to SGLT-2 inhibitors.

The committee examined evidence from the FLOW trial. While the trial identified clinically important benefits in terms of renal protection and glycaemia, subcutaneous semaglutide was not cost effective in the economic model. Therefore, the committee did not recommend semaglutide, any other GLP-1 receptor agonists or tirzepatide for this population.

eGFR above 20 ml/min/1.73 m

and up to 30 ml/min/1.73 m

People with an eGFR below 30 ml/min/1.73 m

cannot take metformin. However, the committee agreed that people with an eGFR above 20 ml/min/1.73 m

could still be offered an SGLT-2 inhibitor to reduce the risk of end-stage renal events and cardiovascular events (including cardiovascular mortality, myocardial infarctions, non-fatal strokes and hospitalisations for heart failure) from type 2 diabetes. The committee recommended dapagliflozin and empagliflozin because these are the 2 SGLT-2

Danh mục: medication

inhibitors that are licensed for use in this population. The committee noted that while the cardiovascular and renal protection provided by SGLT-2 inhibitors are retained at eGFR values below 30 ml/min/1.73 m

, the glycaemic benefits may reduce or be absent. Therefore, adding a DPP-4 inhibitor was recommended, because it is effective at reducing HbA1c and relatively safe for people with an eGFR between 20 ml/min/1.73 m

and 30 ml/min/1.73 m

Data on health inequalities showed that people living in the most deprived areas experience the greatest benefits for their health from SGLT-2 inhibitors. The committee believe this is an important reason for ensuring universal access to SGLT-2 inhibitors.

eGFR below 20 ml/min/1.73 m

DPP-4 inhibitors are effective at reducing HbA1c and have fewer adverse effects than other comparable options. If

DPP-4 inhibitors are contraindicated, not tolerated or not effective for people with an eGFR below 20 ml/min/1.73 m

, then in the committee's experience the best option is either pioglitazone or an insulin-based treatment. The committee acknowledged that a sulfonylurea could increase the risk of hypoglycaemia as renal impairment increases, so would not be a good option for this group. Pioglitazone might worsen cardiovascular outcomes and should be avoided for people with heart failure.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendations might affect practice

Metformin

The recommendations about the use of metformin may lead to a change in current practice but should not lead to a significant cost or resource impact. The price of modified-release metformin can fluctuate but, in December 2025, was lower than the cost of standard-release metformin.

SGLT-2 inhibitors

The recommendations about the use of SGLT-2 inhibitors should not lead to a change in current practice.

DPP-4 inhibitors, pioglitazone and insulin

The recommendation about the use of DPP-4 inhibitors, pioglitazone and insulin should not reflect a change in current practice.

1.19 People with frailty

1.19.1

For adults with type 2 diabetes and frailty:

Offer modified-release metformin.

Only offer an SGLT2 inhibitor if the person's level of frailty does not place them at risk of adverse events from such a medicine (for example, volume depletion or hypotension).

[2026]

1.19.2

If metformin is contraindicated or not tolerated, assess whether their level of frailty places the person at risk of adverse events from SGLT-2 inhibitors:

if it does not, consider monotherapy with a SGLT-2 inhibitor

if it does, consider monotherapy with a DPP-4 inhibitor.

[2026]

1.19.3

Danh mục: medication

Consider reviewing the person's overall diabetes treatment plan to ensure that they are taking the smallest effective number of medications, at the lowest effective dosage.

For further guidance on making a treatment plan for people with frailty (including when associated with multimorbidity), see

NICE's guidelines on clinically assessing and managing multimorbidity

medicines optimisation

shared decision making

[2026]

Why the committee made these recommendations

Because of concerns about adverse effects and polypharmacy, the committee agreed that SGLT-2 inhibitors may not be appropriate for some people with frailty and type 2 diabetes.

There was no specific evidence for people with frailty, so the committee could not recommend a particular method of assessment or cutoff for prescribing SGLT-2 inhibitors. The decision would need to be made based on clinical judgement, taking into account the needs of each person.

The committee recommended medicines for this group based on:

their own expertise

common medicine contraindications in this group

their knowledge of which medicines were likely to have the most manageable side effects.

The committee did not recommend GLP-1 receptor agonists and tirzepatide for people with frailty. However, they agreed that there is no additional safety risk for this population. Therefore, if a person has a relevant indication and frailty, they can still be offered a GLP-1 receptor agonists or tirzepatide.

Full details of the evidence and the committee's discussion are in:

evidence review E: initial management

evidence review F: subsequent management

How the recommendations might affect practice

Metformin

The recommendations about the use of metformin may lead to a change in current practice but should not lead to a significant cost or resource impact. The price of modified-release metformin can fluctuate but, in December 2025, was lower than the cost of standard-release metformin.

DPP-4 inhibitors

The recommendations about the use of DPP-4 inhibitors should not reflect a change in current practice.